

CS 6375

ASSIGNMENT 2

Names of students in your group:

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Number of free late days used: __0__

Note: You are allowed a **total** of 4 free late days for the **entire semester**. You can use at most 2 for each assignment. After that, there will be a penalty of 10% for each late day.

Please list clearly all the sources/references that you have used in this assignment.

1.1 Gradient Descent Derivation

Let each transformed input be $z_i = x_i + x_i^2$, so the unit output is:

$$O = w_0 + w_1 z_1 + w_2 z_2 + \dots + w_n z_n = w_0 + \sum_i w_i z_i$$

Step 1: Define the error(squared error, standard convention with $\frac{1}{2}$ for clean derivative):

$$E = \frac{1}{2}(t - o)^2$$

Step 2: Take the gradient with respect to each weight w_i ($i \geq 1$):

$$\partial E / \partial w_i = \partial E / \partial o \cdot \partial o / \partial w_i$$

- $\partial E / \partial o = -(t - o)$
- $\partial o / \partial w_i = z_i = (x_i + x_i^2)$

$$\text{so: } \partial E / \partial w_i = -(t - o) (x_i + x_i^2)$$

Step 3: Gradient with respect to bias w_0 :

$$w_i \leftarrow w_i + \eta (t - o) (x_i + x_i^2) \text{ for } i = 1 \dots n$$

$$w_0 \leftarrow w_0 + \eta (t - o)$$

1.2a – Output in terms of weights, inputs, general $h(x)$

From the diagram: node 3 receives from x_1 (w_{31}) and x_2 (w_{32}); node 4 receives from x_1 (w_{41}) and x_2 (w_{42}); node 5 combines both.

- Hidden node 3: $h(w_{31} \cdot x_1 + w_{32} \cdot x_2)$
- Hidden node 4: $h(w_{41} \cdot x_1 + w_{42} \cdot x_2)$
- Output: $y_5 = h(w_{53} \cdot h(w_{31} \cdot x_1 + w_{32} \cdot x_2) + w_{54} \cdot h(w_{41} \cdot x_1 + w_{42} \cdot x_2))$

1.2b – Vector notation

With $X = [x_1, x_2]^T$, $W(1) = [[w_{31}, w_{32}], [w_{41}, w_{42}]]$, $W(2) = [w_{53}, w_{54}]$:

$y_5 = h(W(2) \cdot h(W(1) \cdot X))$ where h is applied elementwise to the hidden layer's pre-activation vector $W(1) \cdot X$

1.2c – Showing sigmoid and tanh nets can represent the same function

Step 1- Find the relationship between h_s and h_t .

Start from tanh: $ht(x) = (e^x - e^{-x})/(e^x + e^{-x})$

Multiply numerator and denominator by e^{-x} instead, use the known algebraic identity – divide through and rearrange:

$$ht(x) = 1 - 2/(e^{2x} + 1)$$

Compare to sigmoid: $hs(2x) = 1/(1 + e^{-2x}) = e^{2x}/(e^{2x} + 1)$

$$\text{So, } 2 \cdot hs(2x) - 1 = 2 \cdot [e^{2x}/(e^{2x} + 1)] - 1 = [2e^{2x} - e^{2x} - 1]/(e^{2x} + 1) = (e^{2x} - 1)/(e^{2x} + 1)$$

Multiplying numerator/denominator by e^{-x} confirms this equals $(e^x - e^{-x})/(e^x + e^{-x}) = ht(x)$. So:

$$ht(x) = 2 \cdot hs(2x) - 1$$

Step 2 – to show the networks are equivalent up to linear transformation of weights.

Since $ht(z)$ is just an affine transform of $hs(2z)$, any tanh unit with pre-activation $z = w \cdot x$ can be rewritten as:

$$ht(w \cdot x) = 2 \cdot hs(2w \cdot x) - 1 = 2 \cdot hs((2w) \cdot x) - 1$$

This means, a tanh-activated unit with weight vector w is equivalent to a sigmoid-activated unit with weight vector $2w$, followed by the affine transform ($2 * \text{output} - 1$).

Because this affine transform (scaled by 2, shift by -1) is linear, it can be absorbed into the next layer's weights and biases rather than requiring a nonlinear change. So a full network built with tanh throughout can be converted to an equivalent sigmoid network by:

- Doubling every weight feeding into a hidden/output unit ($w \rightarrow 2w$), and
- Adjusting the next layer's incoming weights and bias to absorb the resulting $* 2$ and -1 factors from each layer below it.